

Supplementary Information: Active SolvAI

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1 Scope and target convention

The simulation-reconstruction target and experimental hydration-free-energy target are distinct. Raw Arbalest annihilation $dH/d\lambda$ is retained with its native sign; a hydration free-energy integral uses the documented sign conversion.

2 Identity and exposure audit

The Phase 1 cohort contains the 72 ARROW molecules for which all three PIMD2 probe windows completed. Molecule identity is fixed by the parent benchmark InChIKey and connectivity key; no molecule was removed according to its response or endpoint error. These molecules are development-exposed through the parent SolvAI evaluation and are not an external validation cohort.

3 Parent SolvAI reproducibility

Parent metrics, model hashes and smoke predictions are regenerated by the release reproduction script.

4 Response inventory

The three campaigns attempted 76 molecules at each of nominal $\lambda = 0.1, 0.5$ and 0.9 . Successful counts were 72, 74 and 73, respectively; 72 molecules completed all three. Each successful window used two beads, 500 equilibration steps and 2,500 production steps at 0.002 ps per step. Sequential prefix summaries used the first 10%, 20%, 40%, 70% and 100% of stored frames.

5 Phase 1 model matrix

All one-, two- and three-point subsets of nominal $\lambda = \{0.1, 0.5, 0.9\}$ are predeclared. Raw actual observations, actual-minus-predicted residuals, shuffled observations and simple generic controls are compared on identical molecules and outer splits. The primary correction was a standardized ridge model whose regularization parameter was selected only within the outer training data. It used the total SYSTEM $dH/d\lambda$ residual at all three points. The matched frozen SolvAI baseline was reconstructed with the same 1,280 public labels, outer-training ARROW labels, sample weights, corrected response teachers and three-member ExtraTrees endpoint ensemble as the parent confirmation.

Across five partitions the frozen baseline MAE was 0.1864 and the residual-corrected MAE was 0.1899 kcal mol⁻¹. The candidate-minus-baseline paired change was +0.00348 (95% bootstrap

interval +0.00096 to +0.00609). The matched predicted-response, raw-response and mean shuffled-residual MAEs were 0.1916, 0.1907 and 0.1893 kcal mol⁻¹, respectively.

6 Sparse posterior diagnostic

Generic and molecule-conditioned Gaussian models were fitted to the three response coordinates with Ledoit–Wolf covariance shrinkage. Held-out observations were conditioned using their five-contiguous-block standard errors. Molecule-conditioned interpolation improved several one- and two-point hidden-coordinate tests, but hidden-point MAE remained at least 2.4 kcal mol⁻¹ and the intervals were wide. This is not a dense-curve validation and cannot establish accurate thermodynamic integration.

7 Prospective dense-sentinel protocol

The dense protocol was frozen at Git commit 512aad3. Four calibration molecules (propane, ethanol, acetone and pyridine) and eight prospective sentinels (cyclohexane, octane, octanol, acetic acid, acetamide, dimethyl ether, ethyl acetate and anthracene) were fixed without reference to ungenerated dense responses. All molecules were already exposed through parent-project development; prospectivity refers only to the twelve newly generated response windows per molecule.

Native Arbalest used force field 933, a 24-Å explicit-water periodic box, 298 K, 1 bar, two PIMD beads, PILE thermostating and the inherited Berendsen production barostat. Each window used a 0.002-ps step, 250 minimization steps, 500 equilibration steps and 2,500 production steps. The fixed 15-point grid was $\lambda = \{0, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1\}$. Existing observations at 0.1, 0.5 and 0.9 were reused. A new window passed only if Arbalest exited normally, printed its success marker, produced one finite SYSTEM energy series with at least 45 frames, and had mean temperature in 250–350 K and density in 0.70–1.30 g cm⁻³.

The generic prior was the calibration-set mean dense curve. The structure-conditioned prior was the target molecule’s frozen, molecule-held-out PIMD2 response prediction at 0.1, 0.5 and 0.9, extended to the grid by shape-preserving interpolation. Calibration-only leave-one-molecule-out negative log predictive density selected the squared-exponential length scale, residual amplitude and observation-noise inflation. Candidate-window noise used only calibration-set standard errors; the measured target-window error entered the posterior only after acquisition.

Budgets contained 3, 4, 5, 7, 9, 12 or 15 total windows, including the inherited three. Comparators were fixed, uniform-maximin, 20 random schedules, curvature-only, generic Bayesian quadrature, direct interpolation and a non-deployable oracle. The primary decision used budgets five and seven. A positive result required at least 0.10 kcal mol⁻¹ improvement over the strongest deployable comparator, a 90% paired-bootstrap upper bound below zero, improvement on at least six of eight molecules and 75–100% interval coverage; or at least 20% fewer windows to reach 0.20 kcal mol⁻¹ with the same coverage requirement.

Table 1: Frozen primary prospective comparisons. Positive differences favour the comparator.

Windows	Active MAE	Comparator MAE	Difference	90% interval	Improved
5	1.701	1.153	+0.548	[−0.101, 1.159]	2/8
7	1.608	1.092	+0.516	[−0.392, 1.491]	3/8

Nominal 90% coverage was 0.750 and 0.875 at five and seven windows; mean interval widths

were 5.406 and 4.893 kcal mol⁻¹. No Bayesian method reached the predeclared stopping half-width of 0.10 kcal mol⁻¹. The non-deployable oracle obtained same-curve MAEs of 0.337 and 0.068 kcal mol⁻¹ at the same budgets. Because it selected and scored windows against the same finite trajectory, these values are optimistically selected diagnostics rather than independent evidence of placement headroom.

8 Independent-noise oracle audit

The post-hoc diagnostic protocol was committed before cross-block scoring at Git commit `d0167dc`. Every 51-frame trajectory was divided into four contiguous blocks; all six ordered non-overlapping two-block selection/two-block evaluation assignments were used. The cross-fitted oracle selected locations from the selection blocks, populated them using evaluation-block responses and was scored only against the evaluation-block dense integral. At five and seven windows it was worse than uniform direct integration by 0.431 and 0.409 kcal mol⁻¹, respectively. Mean reversal-pair Jaccard similarities of added locations were 0.139 and 0.238. Complementary-half dense integrals differed by a mean of 1.866 kcal mol⁻¹ and a median of 1.382 kcal mol⁻¹. The frozen conclusion was therefore inconclusive owing to response noise; the diagnostic removed the original oracle-headroom interpretation without changing the deployable-policy no-go.

9 Independent-replica power gate

The proposed resolution experiment comprised the same eight sentinels, all 15 lambda windows and two independently initialized 50-ps PIMD2 replicas. Its protocol and decision rules were committed at Git commit `52827f8` before the formal power calculation. One hundred thousand Monte Carlo experiments retained the observed centred molecule-effect pattern, imposed a mean oracle advantage of 0.30 kcal mol⁻¹ and scaled the existing 2.5-ps residual variance by inverse production time. A launch required at least 0.80 power at both five and seven windows and at least 0.80 probability that independent dense integrals would pass the registered 0.50-kcal mol⁻¹ mean and 0.30-kcal mol⁻¹ median discrepancy thresholds.

Projected power was 0.00000 at five windows and 0.00001 at seven; dense-reference reliability probability was 0.44602. The campaign was stopped before simulation, avoiding 12 ns of PIMD2 production and an estimated 32.44 production GPU-hours. The low power reflected both finite-trajectory noise and empirical heterogeneity: after shifting the mean effect to -0.30 kcal mol⁻¹, only four of eight molecules at five windows and five of eight at seven favored oracle placement. A planning-only mean-effect study would require about 105 molecules, two replicas, 15 windows and 75 ps per replica (638.68 projected production GPU-hours), but would not establish the registered cross-molecule consistency claim and was not authorized.

10 Compute accounting

The new campaign generated 48 calibration and 96 prospective windows. All 144 passed quality control on their first attempt. Production totaled 720 ps, 288 bead-windows and 900,000 nominal bead-steps. Measured simulation wall time was 6,954.3 s and GPU accounting was 1.932 GPU-h. No failed-window cost was incurred. Exact fast/slow force-kernel evaluations are not emitted by Arbalest and are therefore not reported.

11 Failure attribution

The endpoint null and shuffled-response control support absence of usable endpoint residual signal for the tested protocol. The dense response standard error was material (mean 2.283, median 1.861 and maximum 15.087 kcal mol⁻¹). Same-curve oracle replay was strongly optimistic and did not independently establish molecule-specific placement headroom. The longer-replica proposal was stopped before simulation because its prospectively calculated power and dense-reference reliability were inadequate. Cross-fidelity escalation was not attempted because the same-fidelity prerequisite failed. Hamiltonian bias cannot explain error against the same-Hamiltonian dense reference, but may limit transfer to experiment. No implementation failure was detected.

12 Negative results

Direction A failed the nested endpoint gate. Direction B failed its prospective same-Hamiltonian criterion. Direction C was prospectively contingent on Direction B and was not launched. The apparent same-curve oracle headroom did not survive held-out-block scoring, and the proposed independent-replica resolution experiment failed its prospective power gate before simulation. Tier-B remained unopened. All killed or gated directions are recorded in `NULL_RESULTS.md` and the append-only experiment ledger.